

Sarthak Sahoo

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RESEARCH INTERESTS

System Dynamics and Control, Mechatronics, Motion Planning, Design and Manufacturing

EDUCATION

University of Michigan, Ann Arbor, Michigan, USA

Aug 2026 – May 2028

Master of Science in Robotics

National Institute of Technology Rourkela, Odisha, India

Nov 2020 – Jun 2024

Bachelor of Technology in Mechanical Engineering

Cumulative GPA: **3.81/4.00**

Bachelors Project: *Fabrication, modelling and control of cable-driven parallel robot*

SKILLS

- **Hardware:** NI myRIO/cRIO, Arduino
- **CAD and Design:** SolidWorks, Siemens NX, AutoCAD, CATIA, Simscape Multibody
- **Manufacturing and Analysis:** GD&T, DFMA, engineering drawings, ANSYS, 3D printing, laser cutting
- **Programming:** Python, C++, MATLAB, LabVIEW, Java
- **Productivity Tools:** $\LaTeX$, Microsoft PowerPoint, Word, Excel; Google Docs, Sheets, Slides, Drive

PATENTS & PUBLICATIONS

1. Sorathiya, D., **Sahoo, S.**, & Natarajan, V. (2025). *Design, fabrication and control of a cable-driven parallel robot*. (Experimental Abstract in the Proceedings of the 10th Indian Control Conference (ICC), IISER Bhopal, India, December 9–11, 2024.) [[arxiv](#)][[PDF](#)]
2. Anup Nandy, Sougatamoy Biswas, Asim Kumar Naskar and **Sarthak Sahoo**, Title: *Assistive Robotic System and Method for enhancing social skills in user with autism spectrum disorder*, Patent No: 579221, Indian Patent Office, Granted: 29/01/2025. [[patent](#)]

RESEARCH EXPERIENCE

Continuum Dynamics & Control (CDC) Lab | IIT Bombay | [[Report](#)]

Mumbai, India

Role: Junior Research Fellow, Guide: Prof. Vivek Natarajan

Feb 2025 – Jul 2026

- Formulated the mathematical model of a coupled rigid-flexible system using Hamilton's variational principle, considering both rigid-body motion and elastic deformation.
- Developed a MATLAB-based FEM model for an 1-D Euler-Bernoulli beam using Galerkin method and Hermite elements; predicting natural frequencies within $\pm 0.1\%$ error, & estimating modal participation factors from experimental data.
- Estimated elastic and damping properties of 1-D flexible beams of unknown steel grade from experimental vibration data using Micro-Epsilon **laser displacement sensors**, **MPU6050 IMUs**, **FFT-based natural-frequency estimation**.
- Integrated an ABZ optical encoder with NI myRIO and LabVIEW FPGA for real-time position measurement and closed-loop feedback control of a single-axis linear stage used in an inverted-pendulum experimental setup.
- **Stacks:** *Python (NumPy, SciPy)*, *MATLAB*, *myRIO*, *LabVIEW*, $\LaTeX$

Continuum Dynamics & Control (CDC) Lab | IIT Bombay | [[Poster](#)]

Mumbai, India

Role: Project Research Intern, IRCC Research Fellowship, Guide: Prof. Vivek Natarajan

Jan 2024 – Jun 2024

- Designed and fabricated an underactuated cable-driven parallel robot with optimized workspace and actuator layout, achieving ± 5 mm **trajectory-tracking** repeatability, using closed-loop steppers and Dyneema cables.
- Developed a custom 3D-printed winch drum and flexible coupling for slip- and backlash-free cable actuation; sized and selected the motor based on cable tension and 2–300 RPM speed constraints.
- Developed a LabVIEW FPGA code to generate PFM signals for synchronized position control of 3 actuators
- Tested motor accuracy using Arduino/Python DAQ with incremental encoder feedback to validate step, ramp, and sinusoidal velocity responses.

- **Stacks:** LabVIEW, SolidWorks, Arduino, UltiMaker Cura

Machine Intelligence & Robotics (MIRO) Lab | NIT Rourkela | [CAD] Rourkela, India (Remote)

Role: Undergraduate Research Assistant, Guide: Prof. Anup Nandy Nov 2023 – Jun 2024

- Designed and prototyped a wheeled dual-arm mobile social robot, considering user height and bot stability; integrated wheel motors for omnidirectional movement and arm motors for **2-DOF manipulation via UART protocol**.
- Modeled the 2-DOF arm in MATLAB Simscape Multibody; derived kinematics using **geometric method** and dynamics using **Euler-Lagrange method** and validated the model.
- The uniqueness of this approach in addressing children with autism spectrum disorder led to a filed and published Indian patent (2025).
- **Stacks:** SolidWorks, Arduino, MATLAB.

INDUSTRIAL EXPERIENCE

Bajaj Auto Limited | [Slides]

Pune, India

Role: Graduate Trainee Engineer (Manufacturing Engg.), Mentor: Rajesh S Mangrulkar Aug 2024 – Dec 2024

- Extended tool life of a spring-plunger inductive proximity sensor poka-yoke for Woodruff key detection from **1 to 4 days** by reducing spring stiffness and optimizing plunger geometry.
- Improved inspection reliability of a valve collet press system by **20%** through systematic root-cause analysis, functional GD&T drawing studies, **SPC on 200 samples**, and laser sensor recalibration, applying TPM and quality tools throughout.
- Optimized fuel leak testing rigs by refining pneumatic control logic and pressure settling time, reducing cycle time **from 21 s to 17 s** and enhancing line productivity.
- **Stacks:** MS Office, QC charts, Data interpretation

Bajaj Auto Limited | [Slides]

Pune, India

Role: Summer Octane Intern (Manufacturing Engineering), Mentor: Rajesh S Mangrulkar May 2023 – Jul 2023

- Evaluated **20+ Poka-yoke mechanisms** & performed material and information flow mapping across **2 Pulsar assembly conveyors** and documented inspection logic, to improve error-proofing traceability on the MES platform.
- Standardized **15+ UI dashboards** for digital production boards and line-stage displays, enabling real-time quality monitoring and reducing manual inspection delays by **10 seconds**.
- Recognized with a **Pre-Placement Offer (PPO)** for outstanding performance and contributions to digital manufacturing initiatives.
- **Stacks:** MS Office, QC charts, Data interpretation

PROJECTS

Design & fabrication of an Autonomous Underwater Vehicle (AUV) | [CAD]

Rourkela, India

Mechanical Subsystem Lead, Team Tiburon (AUVNITRKL) Dec 2021 – Jan 2024

- Designed a modular frame for an intelligent underwater vehicle using 3D CAD modeling software, considering strategic placement and orientation of 8 BlueRobotics T100 thrusters improving maneuverability from **4-DOF to 6-DOF**.
- Improved the bot weight from **21 kg to 20.28 kg** by performing structural optimization and employing lightweight modular aluminum extrusions.
- Verified structural integrity and safety factors of the hulls for submerged operation at depths ranging from **0.5 m to 1.0 m** using hydrostatic structural analysis.

Design & development of a e-Human Powered Vehicle with landing gear | [Report]

Rourkela, India

Captain, Team BlueStreak, ASME NIT Rourkela Student Team May 2021 – Apr 2023

- Designed a semi-recumbent 2-wheeler chassis; along with ergonomic analysis on CATIA for **5th and 50th percentile** manikins to optimize rider posture, vision window, and joint kinematics.
- Conducted static loading and modal analyses of the rollover protection structure in ANSYS; optimized triangulation geometry to decrease deformation **from 0.10 mm to 0.07 mm (30% reduction)**, to meet with competition objectives.
- Developed and simulated a landing-gear mechanism in Simscape Multibody, achieving controlled deployment within **8 s**, improving vehicle stability during starts and stops.
- Computed motor traction parameters and battery sizing in MATLAB/Simulink for a 3-wheeled recumbent design.

AWARDS

- **Runners Up @ Bajaj Auto OHM Challenge 2024** | [Slides_Prelims] | [Slides_Finale] Mar 2024
Led team QO9D which secured **National Runners-Up & Campus Winner** in the OHM Challenge 2024, organized by Bajaj Auto & Chetak Technology, among **1500+ teams** for proposing a "control algorithm enabling swappable EV battery packs".
- **IRCC Research Internship Awards 2024** Dec 2023
Secured a **6-month funded research internship** in the Department of Systems & Control Engineering, IIT Bombay, under the IRCC Research Fellowship (2023–24), granted to students in the **top 10 percentile** of their class.
- **National Runners Up @ ASME e-HPVC 2023** Apr 2023
Team BlueStreak 10.0 (under my leadership) secured **National Runners-Up** in Critical Design Report Presentation at the ASME e-HPVC India.
- **Proof of Concept @ L&T Techgium** Mar 2024
Qualified POC stage for proposing an "Active Torque Vectoring System using predictive error-based control" at 6th edition of L&T Techgium.
- **National Runners Up @ DecodeEV Hackathon 2022** | [Presentation] Sep 2022
Team HACKOHOLICS secured **National Runners-Up among 100+ teams**, for an AI-based solution to prevent EV thermal runaway at the DecodeEV Hackathon 2022 organized by WUElev8 & IIIT Delhi.
- **Global rank 2nd @ ASME e-HPVC 2022** | [Slides] Apr 2022
Team BlueStreak 9.0 ranked **2nd in overall design and 3rd in innovation** all over the world in the e-HPVC competition organized by American Society of Mechanical Engineers (ASME).
- **National Anveshika Experimental Skill Test 2020** Aug 2020
Awarded with Excellent Grade in the national-level experimental physics competition, National Anveshika Experimental Skill Test (NAEST), conducted by CCE IIT Kanpur and Indian Association of Physics Teachers (IAPT).
- **Recognition for violin performance at ASME e-FEST Digital** Apr 2021
Recognized for outstanding performance in Hindustani classical Violin at the ASME E-FEST Digital 2021.
- **Sangeet Bhushan in Hindustani Classical Violin** Jan 2022
Completed Sangeet Bhushan Final exam in Violin (Instrumental Classical) with Distinction in practical, granted by Pracheen Kala Kendra, Chandigarh, India.

LEADERSHIP, TUTORING & VOLUNTEERING

Career Development Centre, NIT Rourkela | [certificate] Rourkela, India
Training Coordinator Dec 2023 – Jun 2024

- Coordinated with alumni across automotive, power & heavy engineering, oil & gas, and FMCG sectors to organize **5 career-development webinars** for sophomore and junior Mechanical Engineering students.
- Designed and administered online assessments for **250+ students**, developing question banks spanning quantitative aptitude, logical reasoning, and core Mechanical Engineering subjects.
- Dedicated **5–6 hours weekly** to mentoring and assessing students through **40+ one-on-one mock interviews**, resume reviews, and personalized feedback sessions, while curating learning resources and certification pathways to enhance placement readiness.

Institute Counselling Services, NIT Rourkela Rourkela, India
Student Mentor, Mechanical Engineering May 2022 – Apr 2023

- Served as a mentor for **10 freshman Mechanical Engineering students**, providing academic, socio-cultural, and personal support to ease their transition into university and hostel life.
- Supported my mentees in the **PH1001 Engineering Physics** course by clarifying concepts in foundational physics, relativistic mechanics, and introductory quantum mechanics, while sharing supplementary learning resources.

Team BlueStreak, ASME NIT Rourkela Student Section Rourkela, India
Captain & Powertrain subsystem member May 2022 – Apr 2023

- Led and managed a **25-member** interdisciplinary team for ASME e-HPVC India 2023, overseeing vehicle design, manufacturing, and parts procurement, while conducting training sessions for **30+ new recruits** to ensure effective knowledge transfer and timely competition delivery.

HeartBeats, Fusion Music Band of NIT Rourkela

Violinist and Vocalist

Rourkela, India

Jan 2021 – Apr 2023

- Performed at 4th National Level Shashtriya Sangeet Samaroha; ASME E-FEST digital 2021, MAKTUB 2022 (NITRUTSAV 2022) and DHWANI 2021 (classical music festival of NIT Rourkela).

SELECTED COURSES

Bachelor's Courses

- Mechanical vibrations
- System dynamics and control
- Mathematical methods
- Mechatronics
- Dynamics of machines

Additional courses & Certifications

- Vibration and structural dynamics (IIT Kharagpur, NPTEL) [**credential**]
- Machine dynamics with MATLAB (RWTH Aachen University, edX) [**credential**]
- Certified SolidWorks Professional in Mechanical Design (Dassault Systemes) [**credential**]
- MATLAB Fundamentals – MathWorks [**credential**]
- Control System Design with MATLAB and Simulink (MathWorks) [**credential**]

ENGLISH & GRE TESTS

TOEFL (iBT): 112 (overall score)

Listening: 28 | Reading: 29

Speaking: 27 | Writing: 28

Test date: October 2025

GRE General Test:

Quant: 168 | Verbal: 146

Analytical writing: 4.0

Test date: November 2025